

Rational Unified Process (RUP)

Full software project life cycle is divided into four parts, Phases. Different types of works included here.

- Modeling
- Analysis
- Design
- Implementation
- Testing
- Application

Key Phases of RUP:

1. Inception: Defines the project

- scope (Project idea and limitation)
- core requirements
- feasibility

2. Elaboration

- Domain Analysis
- Architecture
- Risk

Example:

Payment system কাজ করবে,

Security কি আছে?

Data base থেকে user handle করতে পারবে?

3. Construction

- Coding

- Component Development

- Testing

Mainly software build in this phase.

4. Transition: Mainly software delivered to the user in this step.

- Final Testing

- Deployment

- Training

Advantages of RUP:

- It provides good documentation.
- It provides risk management support.
- It reuses the components, and hence total time duration is less.
- Good online support is available in the form of tutorials and training.

Disadvantages of RUP

- Requires expert professionals
- Complex process
- Hard to integrate again and again.
- High Dependency on risk management.

When to use RUP

- Large and complex software
 - Banking system
 - Hospital system
 - Enterprise system
- Customized software development process
- Incremental and iterative development
(Software विकास improve करना रहता)
- Multiple teams/stakeholders

- Strong documentation / modeling

- Development process tailor

(project/environment based information विद्युत
process adjust करके रखने)

When not to use RUP

- Project small / simple

- Timeline small

- Stakeholder needs lightweight or fast
development methods

- Small Team

- Team is not experienced with structured
process (specially for beginner team members)

- Team prefers highly ~~flex~~ flexible / informal
Agile approach (They don't like heavy
documentation)

- If too much documentation not needed, then
avoid RUP

Proto type Methodology

Instead of developing full-fledged software, the ~~proto type version of the final~~ model allows developers to work on the proto type version of the final product.

Customer see proto type → evaluate → give feedback

Based on the gathered feedback, the proto type goes through several iterations of refinement ~~and~~ until it's deemed satisfactory by the customer.

Advantages of Proto type

1. ~~Early~~ Early problem Detection (The proto type helps identify issues before the real software is developed, reducing the risk of product failure.

2. Customer satisfaction - Customers can test the prototype and give feedback, so the final product matches their needs.

3. Regular discussions and feedback build trust and good communication with the customer.

(Better customer relationship)

4. Clear requirements - The prototype helps collect detailed information and requirements, which are used to build the final version.

Disadvantages of Prototype

- Development may be delayed - Too much testing and feedback with the customer can increase the project.

- Customer expectations may differ - The customer may think the prototype is the final product, leading to unrealistic expectations.

- Risk of cost overrun - Building and improving the prototype often costs extra money, which may increase overall project cost

When to use prototype

- Requirements are not clear in the beginning,
- Early user feedback is needed
- The system has a complex user Interface (UI)
- Developers need to demonstrate the system to stakeholders
- The project involves high user interaction
- The team is exploring new ideas or innovative features
- Requirements need validation before full development.

When not to use Prototype

- Requirements are clear and stable
- The system is simple and straightforward
- The project has strict deadlines
- Users may confuse the prototype

with the final product.

- The project needs strong documentation

(IV) and strict processes.

Scrum Methodology

Scrum is arguably one of the most flexible software development methodologies available. Scrum is an Agile-based methodology. Scrum depends on sprint.

Scrum is :

- flexible
- Incremental
- Iterative

Three main roles of Scrum:

1. Product owner

2. Scrum Master

3. Development Team

1. Product Owner: Product owner decides what to build.

- Represents the customer
- Collects client requirements and feedback

- Decides what features should be built

the first

- Ensures the team delivers value to the customer

2. Scrum Master: Scrum Master helps the team work smoothly.

- Acts as the facilitator of the Scrum process

- Helps the team follow Scrum rules.

- Removes obstacles that slow the team.

- Supports communication and teamwork.

3. Development Team: Development Team

builds the software.

- Designs, codes, tests, and delivers the software.

- Works together to complete tasks in each sprint.

- Builds the actual product.

Shortcut:

Role

Responsibility

Product Owner

Defines requirements & priorities

Scrum Master

Manage the scrum process

Development Team

Develops and tests the product.

Sprint

Sprint is a short development cycle.

A sprint consist, upto 4 weeks.

Goal = Build and deliver a working part of the product.

At the end of every sprint = Test the feature, fix issues, and collect customer feedback.

Advantages of Scrum:

WBS

Rules for generating a WBS

- Decomposition: একটি বড় project কে ছোট ছোট
- ~~Hierarchy~~ Hierarchy: সাজানো জায়গা করে decomposition বসানো.
- Hierarchical structure: কাজগুলোকে Level 1, Level 2, Level 3 আকারে আলাদা করা
- Clear Deliverables: প্রতিটি module এ একটি নির্দিষ্ট ফলাফল (output) থাকবে।
- Mutually Exclusive: একই কাজ ২ জায়গায় থাকবে না।

Wrong X

- User Registration
- Student

Correct

- Student Registration
- Teacher Registration

5. Consistent Numbering: Use a logical numbering system to identify hierarchy.

level 1 & code: 1000

6. Clarity: Task names should be simple, specific, and easy to understand.

7. Activity Breakdown: Break complex tasks into smaller work packages for better estimation.

Estimating Duration for Two/Three major WBS

components:

subtasks scenario based

WBS code	Task Name	Sub tasks	Duration
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→ Project Management

→ System Design

→ Development

→ Testing

→ Deployment

→ Maintenance.